

Guide to Interpreting the Output of the Fourier Analysis Code (Version 9)

1 Purpose of the code

The v.9 code takes as input a list of pairs (n, μ_n) , where μ_n is the minimum modulus among the eigenvalues of a certain matrix associated to the index n . Its purpose is to determine whether the sequence $\{\mu_n\}$ oscillates in a nearly periodic fashion and, if so, to estimate the period.

Version 9 extends version 8 by adding *envelope normalization* (Section 5), which enables the code to detect periodicity in signals whose oscillation amplitude grows or decays.

The code produces a text report (described in Sections 4–11) and eight diagnostic plots (described in Section 12).

2 Glossary of abbreviations and acronyms

The following abbreviations appear in the code and its output.

ACF **Autocorrelation function.** Given a sequence $x_1, \dots, x_N$ with mean $\bar{x}$, the (normalized) autocorrelation at lag k is

$$\text{ACF}(k) = \frac{\sum_{i=1}^{N-k} (x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2}.$$

By construction $\text{ACF}(0) = 1$. If the sequence is periodic with period T , then $\text{ACF}(T)$ will be a local maximum close to 1. If the sequence is monotone or trend-like, the ACF decays from 1 toward 0 without oscillating.

AR(1) **First-order autoregressive model.** A simple stochastic model in which each value depends linearly on its predecessor plus noise:

$$x_n = \rho x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2).$$

The parameter ρ (reported in the output as “lag-1 autocorrelation” or simply “rho”) measures how strongly consecutive values are correlated. The AR(1) model serves as a *null hypothesis*: it represents data that has short-range correlation but no true periodicity. A spectral peak is declared significant only if it exceeds what the AR(1) model would produce.

Bonferroni correction

A standard multiple-testing correction. When testing m frequency bins simultaneously, the probability of at least one false positive at level α can be as high as $m\alpha$. The Bonferroni correction replaces α with α/m at each bin, so that the *family-wise* false positive rate stays at α . In the code, m equals the number of frequency bins with period $\leq N/3$.

dB	Decibel. A logarithmic unit used in the spectrogram (Section 12). Power in decibels is $10 \log_{10}(\text{power})$.
Dynamic range	In the envelope normalization report (Section 5), this refers to the ratio of the 90th percentile of the estimated envelope to its 10th percentile. A ratio exceeding 3 triggers normalization.
Envelope	The slowly varying curve that traces the peaks of an oscillating signal. For a signal of the form $A(n) \sin(2\pi n/T)$, the envelope is $ A(n) $. See Section 5.
FFT	Fast Fourier Transform. An algorithm that computes the discrete Fourier transform of a length- N sequence in $O(N \log N)$ operations. The discrete Fourier transform decomposes the sequence into a sum of sinusoidal components at frequencies k/N for $k = 0, 1, \dots, N - 1$.
Hann window	A bell-shaped taper applied to the data before the FFT (Section 6). It reduces spectral leakage but has the side effect of strongly attenuating signals near the Nyquist frequency (period ≈ 2), because the window goes to zero at both endpoints of the frequency range. The ACF-based analysis does not use windowing and is therefore unaffected.
MA(w)	Moving average with window width w. The smoothed value at index i is the arithmetic mean of the w values centered at i . Used in the multi-scale period detection (Section 9) to suppress fine-scale structure before re-measuring the ACF.
Nyquist frequency	The highest frequency that a discretely sampled signal can represent: $f_{\text{Nyq}} = 1/2$ (in units of cycles per sample), corresponding to a period of 2. A period-2 oscillation alternates high–low–high–low and sits exactly at the Nyquist limit. The Hann window attenuates signals near this frequency, which is why the FFT may fail to detect a period-2 oscillation even when the ACF detects it clearly.
p10, p90	The 10th and 90th percentiles of the estimated envelope, used to compute the dynamic range.
SG	Savitzky–Golay filter. A smoothing method that fits a low-degree polynomial (here, cubic) by least squares within a sliding window of specified width, and replaces each data value with the fitted value at that point. Unlike a moving average, it preserves local peaks and inflection points better. Used both in detrending (Section 4, Stage B) and in envelope estimation (Section 5).
N	The number of data points in <code>mins_list</code> .
$N/3$	The maximum period considered credible. A period of T observed in N data points corresponds to N/T full cycles; for $T > N/3$ there are fewer than three full cycles, which is generally insufficient to distinguish true periodicity from a trend artifact.
rho (ρ)	The lag-1 autocorrelation of the analysis signal (after detrending and, if applicable, envelope normalization), used as the parameter of the AR(1) null model.

Values near $+1$ indicate strong positive serial correlation (each value close to its predecessor); values near -1 indicate alternation; values near 0 indicate near-independence.

std dev Standard deviation.

3 Section 1 of the report: data summary

The report begins by printing the range of the index n , the number of data points N , and the range of the minimum moduli μ_n . These are for basic sanity-checking.

4 Section 2 of the report: adaptive detrending

Before searching for periodicity, the code removes any broad non-oscillatory trend from the data, because a monotone growth curve (for instance) would inject spurious low-frequency power into the Fourier transform.

Stage A: polynomial detrend

A degree-2 polynomial (parabola) is fitted to the data by least squares and subtracted. This is gentle enough not to absorb any oscillation whose period is less than $N/3$.

The code then examines the residuals to decide whether Stage A was sufficient. It computes the ACF of the residuals and checks two things:

1. Does the ACF cross zero within the first $N/3$ lags? If so, the residual is oscillatory, which is the desired outcome: the polynomial removed drift without destroying periodicity.
2. Is the lag-1 autocorrelation above 0.85? If the ACF does not cross zero and the lag-1 autocorrelation exceeds 0.85, the residual is still dominated by a trend that the parabola could not capture.

Stage B: Savitzky–Golay filter (conditional)

Stage B is invoked only when Stage A leaves a trend-like residual (no zero crossing *and* lag-1 autocorrelation exceeding 0.85). This occurs, for example, when the data follows a ramp-then-plateau shape that no single polynomial fits well.

The code applies a Savitzky–Golay filter with a cubic polynomial and tries several window widths, selecting the one that minimizes the lag-1 autocorrelation of the resulting residual. The window width is capped at $N/3$ to prevent the smoother from absorbing a genuine oscillation whose period is shorter than the window.

The report prints which detrending method was selected and the standard deviation of the final detrended residuals.

5 Section 2b of the report: envelope normalization (new in v. 9)

The ACF and the AR(1) significance test both assume that the oscillation has roughly constant amplitude across the data. When the amplitude varies strongly—for example, an oscillation that decays as $1/n$ —two problems arise:

- The ACF is dominated by whichever portion of the data has the largest amplitude, producing a misleadingly weak overall correlation.
- If the oscillation alternates in sign (negative lag-1 autocorrelation), the AR(1) null model can absorb the alternation as stochastic noise, raising the significance threshold and masking the true periodicity.

To address this, the code estimates the local amplitude *envelope* of the detrended signal:

1. Compute the absolute values $|d_i|$ of the detrended data.
2. Smooth these absolute values with a Savitzky–Golay filter (quadratic, window width approximately $N/10$) to obtain a slowly varying envelope E_i .
3. Impose a floor on the envelope (at the 5th percentile of the nonzero absolute values) to prevent division by near-zero values.

The code then checks the *dynamic range* of the envelope, defined as the ratio of its 90th percentile to its 10th percentile. If this ratio exceeds 3, normalization is applied: each detrended value is divided by the corresponding envelope value, producing a signal with approximately constant oscillation amplitude. If the ratio is 3 or below, the envelope is judged to be sufficiently flat and no normalization is performed.

When normalization is applied, the report prints the dynamic range and the standard deviation of the normalized signal. When it is not applied, the report notes this and prints the dynamic range for reference.

All subsequent analysis—the FFT, the AR(1) significance test, and the multi-scale ACF—operates on the signal *after* envelope normalization (or on the unmodified detrended signal if normalization was not needed). This signal is referred to internally as the *analysis signal*.

6 Section 3 of the report: windowed FFT

The analysis signal is multiplied by a *Hann window* (a bell-shaped taper that goes to zero at both ends) before computing the FFT. The purpose of windowing is to reduce *spectral leakage*: without it, the FFT implicitly assumes the data is periodic over its full length, and any mismatch at the endpoints spreads spurious power across all frequencies. A multiplicative correction factor compensates for the power lost to the window.

The output of the FFT is a *power spectrum*: for each frequency $f = k/N$, a non-negative number measuring how much of the signal’s variance is attributable to oscillation at that frequency. The corresponding period is $T = 1/f$.

7 Section 4 of the report: significance testing

Not every bump in the power spectrum is meaningful; random noise also produces spectral peaks. To distinguish real periodicity from noise, the code constructs a null model and asks: could the observed peak have arisen by chance?

The AR(1) null model

The null hypothesis is that the analysis signal is generated by a first-order autoregressive process (see the glossary). The theoretical power spectrum of an AR(1) process with lag-1 autocorrelation ρ and variance σ^2 is

$$S(f) = \frac{\sigma^2(1 - \rho^2)}{1 - 2\rho \cos(2\pi f) + \rho^2},$$

following Gilman, Fuglister, and Mitchell (1963). This spectrum is not flat: when $\rho > 0$ it is tilted toward low frequencies (“red noise”), and when $\rho < 0$ it is tilted toward high frequencies. The normalization ensures that the mean of the theoretical spectrum matches the mean of the observed spectrum.

Chi-squared test with Bonferroni correction

Under the null model, the power at each frequency bin is approximately distributed as a scaled chi-squared random variable with 2 degrees of freedom. The 95% and 99% confidence thresholds are computed from this distribution, then inflated by the Bonferroni correction to account for the number of bins tested (see the glossary).

The report prints the number of frequency bins tested and the Bonferroni-corrected per-bin significance levels.

8 Section 5 of the report: significant FFT peaks

The code identifies all local maxima of the power spectrum that exceed the Bonferroni-corrected 95% or 99% confidence threshold, restricted to periods $\leq N/3$. These are listed in a table with columns:

Rank	Ordered by power (strongest first).
Period	$T = 1/f$, measured in index units (i.e., if Period = 60, the pattern repeats every 60 values of n).
Frequency	$f = 1/T$.
Power	The squared magnitude of the Fourier coefficient at this frequency, displayed in scientific notation (e.g., 1.58e+04 means $1.58 \times 10^4 = 15,800$).
Signif.	Whether the peak exceeds the 95% or 99% threshold.

Important caveat. For non-sinusoidal waveforms (sawtooth, square-wave, or similar shapes built from sharp segments), the FFT distributes the oscillation’s energy across many *harmonics* (integer multiples of the fundamental frequency). The highest-power FFT peak may therefore correspond to a harmonic rather than the fundamental period. The multi-scale analysis described next addresses this issue.

9 Section 5b of the report: multi-scale period detection

This section identifies the *fundamental period* by a method that is immune to the harmonic ambiguity of the FFT.

The idea is to examine the analysis signal at progressively coarser scales by smoothing it with moving averages of increasing width $w = 1$ (raw), 3, 5, 7, ... At each scale, the ACF of the smoothed signal is computed, and the first ACF peak above lag w with strength exceeding 0.1 is recorded. (The restriction to lags above w prevents the smoothing itself from creating artificial peaks.)

Different scales reveal different periodicities. At the finest scale (raw data), the ACF detects the shortest repeating feature. At coarser scales, short-period structure is averaged out and longer periods emerge. For instance, a period-2 oscillation produces ACF peaks at lags 2, 4, 6, 8, ... (all integer multiples of 2); the raw scale sees lag 2, while coarser scales see the longer multiples. A sawtooth with period 60 built from teeth with period 3 will show period 3 at fine scales and period 60 at coarser scales.

Identifying the fundamental

The fundamental is *not* simply the longest or shortest detected period. Instead, the code tests each detected period as a candidate fundamental by counting how many of the other detected periods are approximate integer multiples of it (within a 15% tolerance). The candidate that explains the most other periods wins; ties are broken in favor of higher ACF strength.

This logic correctly handles two opposite situations:

- A period-2 oscillation detected at lags 2, 4, 6, 8: period 2 explains all the others as its $2\times$, $3\times$, $4\times$ multiples, so it is identified as the fundamental.
- A sawtooth with period 60 and sub-cycle teeth of period 3: period 3 does not explain period 60 as an integer multiple ($60/3 = 20$, which *is* an integer, so in fact both candidates explain each other; the tie is broken by ACF strength, which will favor the stronger, coarser-scale detection of period 60).

The results are collected into a table with columns:

Period	The lag at which the ACF peaks, in index units.
Scale	Which smoothing was applied. “raw” means no smoothing; “MA(w)” means a moving average of width w .
ACF strength	The value of the normalized ACF at the detected lag. Ranges from -1 to $+1$; values close to $+1$ indicate a strong, clean repetition. Values below 0.2 are considered too weak to be reliable.
Role	The winning candidate is labeled FUNDAMENTAL; all other detected periods are labeled “integer multiple.”

10 Harmonic analysis table

When both a fundamental period and significant FFT peaks are present, the code checks whether each FFT peak is a *harmonic* of the fundamental. The table shows:

$T_{\text{fund}}/T_{\text{peak}}$ The ratio of the fundamental period to the FFT peak’s period. If this is close to a positive integer m , the FFT peak corresponds to the m -th harmonic.

Nearest int The integer nearest to the ratio.

Harmonic? If the ratio is within 15% of an integer, the peak is tagged as “ $\sim mx$ ” (an approximate m -th harmonic). Otherwise it is tagged “independent,” meaning it represents a genuinely distinct period not explained as a harmonic of the fundamental.

A waveform whose FFT peaks are all harmonics of the fundamental is non-sinusoidal but still truly periodic with the fundamental period.

11 Summary

The summary opens with a one-word verdict—**PERIODIC**, **POSSIBLY PERIODIC**, or **NOT PERIODIC**—followed by supporting detail. The verdict is reached by combining the FFT and ACF evidence as follows.

1. **PERIODIC.** Printed when at least one of the following holds:

- the FFT has at least one peak exceeding the Bonferroni-corrected threshold, or
- the multi-scale ACF finds a fundamental with strength above 0.5.

The report then lists the fundamental period, detection method, ACF strength, and any integer multiples detected at other scales. If envelope normalization was applied, this is noted with the dynamic range.

When the FFT and ACF agree, the report says so. When they disagree, it explains why:

- If the ACF is strong but the FFT finds no significant peak, and the fundamental period is close to 2 (the Nyquist limit; see the glossary entry below), the report notes that the Hann window attenuates signals at this frequency. The ACF does not suffer from this limitation.
- For longer periods, it notes that the peak did not clear the Bonferroni threshold but that the ACF strength constitutes strong independent evidence.

2. **POSSIBLY PERIODIC.** Printed when the multi-scale ACF finds a fundamental with strength between 0.2 and 0.5, and the FFT does not confirm. The candidate period is reported, but the reader is advised to interpret with caution.

3. **NOT PERIODIC.** Printed when the ACF strength is below 0.2 or no period is detected at any scale, and the FFT finds no significant peaks.

12 The eight diagnostic plots

1. **Original: Minimum Moduli.** The raw input data (n, μ_n) . Visual inspection of this plot is the most basic sanity check.
2. **Data with Trend.** The raw data (blue) overlaid with the fitted trend (red). Check that the red curve captures the broad shape without tracking the oscillations.

3. **Detrended Residuals.** The data after subtracting the trend, *before* envelope normalization. If detrending worked well, this should oscillate around zero with no visible upward or downward drift. The oscillation amplitude, however, may still vary.
4. **Detrended Data with Estimated Envelope** (or, if normalization was not needed, **Analysis Signal**). When envelope normalization is applied, this plot shows the detrended signal in light blue with the estimated envelope in red (both positive and negative branches). Check that the red curves trace the peaks of the blue oscillation without cutting through them or lagging behind. When normalization was not applied, this plot simply repeats the analysis signal.
5. **Analysis Signal.** The signal that the FFT and ACF actually operate on. If envelope normalization was applied, this is the detrended signal divided by the envelope; the oscillation should now have roughly constant amplitude. If normalization was not applied, this is identical to the detrended residuals.
6. **Power Spectrum with Bonferroni-Corrected Significance.** A log-scale plot of power versus frequency. The dashed curves are:
 - black: the AR(1) red-noise spectrum (the null model),
 - green: the 95% Bonferroni-corrected threshold,
 - red: the 99% Bonferroni-corrected threshold.

Peaks marked with colored dots exceed the corresponding threshold. This plot is the primary evidence for or against periodicity.

7. **Power Spectrum vs. Period.** The same information as plot 6, but with the horizontal axis showing period rather than frequency. This is often easier to read because periods are directly interpretable (“the pattern repeats every T steps”).
8. **Spectrogram.** A time–frequency decomposition showing how the spectral content changes as n increases. The horizontal axis is n , the vertical axis is period, and color represents power in decibels. A horizontal band of warm color indicates a stable periodicity persisting across the data. A band that drifts, appears, or disappears indicates that the periodicity is transient or evolving.

13 Common patterns and their interpretation

Pattern 1: No periodicity (e.g., a monotone growth curve)

The power spectrum shows no peaks above the confidence thresholds. The multi-scale table is empty or reports only weak ACF values. The summary verdict is **NOT PERIODIC**.

Pattern 2: Clean sinusoidal oscillation

A single dominant FFT peak matches the fundamental period from the multi-scale analysis. Harmonics are absent or very weak. The summary verdict is **PERIODIC**, confirmed by both FFT and ACF.

Pattern 3: Non-sinusoidal oscillation (e.g., sawtooth)

The FFT table shows multiple significant peaks. The highest-power FFT peak may have a short period (a high harmonic), but the multi-scale analysis identifies the true fundamental period—which may be either longer or shorter than the dominant FFT peak—by finding the period whose integer multiples best account for all detected periods. The harmonic analysis table shows that the FFT peaks are integer multiples of the fundamental. The summary verdict is **PERIODIC**, and it notes that the dominant FFT peak is a harmonic and that the waveform is non-sinusoidal.

Pattern 4: Periodicity with drift

The detrending selects Stage B (Savitzky–Golay). After detrending, periodicity may or may not be present. The spectrogram can reveal whether the periodicity is stable or confined to part of the data range.

Pattern 5: Amplitude-modulated periodicity

The oscillation is clearly visible in the original data, but its amplitude decays (or grows) across the index range. Without envelope normalization, the ACF would report weak strength and the FFT might fail the significance test. With normalization applied (reported as “Envelope normalization: APPLIED” with a dynamic range exceeding 3), the analysis signal has roughly constant amplitude, and the periodicity is correctly detected. The summary verdict is **PERIODIC** and notes that envelope normalization was applied.

When this pattern occurs, note that the *period* is meaningful but the *power* values in the FFT table refer to the normalized signal, not the original data. The power should be understood as evidence for or against the existence of periodicity, not as a measure of the oscillation’s physical amplitude.

Pattern 6: Very short period with integer-multiple detections

The data oscillates with a short period (e.g., period 2, alternating high and low). The multi-scale table shows detections at several periods that are all integer multiples of the true fundamental: for instance, periods 2, 4, 6, 8 when the true period is 2. These arise because the ACF of a period- T signal has peaks at all lags $T, 2T, 3T, \dots$, and the progressive smoothing shifts which multiple is detected first. The fundamental selection logic correctly identifies the shortest period that explains the others.

The summary verdict is **PERIODIC**, but the FFT section may report no significant peaks. This is not a contradiction: at period 2 (the Nyquist frequency), the Hann window applied before the FFT strongly attenuates the signal. The summary explains this and notes that the ACF is unaffected by the windowing.

This pattern may co-occur with Pattern 5 (amplitude modulation): the Chebyshev polynomial example combines a period-2 oscillation with a decaying envelope.